

Script-Controlled ACL – Restrict Record Access Based on Field Value

6. Project Documentation

6.1 Project Overview

Script-Controlled ACL is a record-level authorization mechanism in which access is controlled by a script that evaluates values stored in the record and attributes of the current user.

6.2 Key Features

- Field-based record restriction.
- Script-controlled authorization.
- Role and department based conditions.
- CRUD-level protection.
- Server-side access validation.
- Audit-friendly access decisions.

6.3 Advantages

- Provides finer control than basic role-only access.
- Can support different business rules.
- Reduces exposure of sensitive records.
- Rules can be changed without redesigning the complete application.

6.4 Limitations

- Poorly written scripts can create security gaps.
- Large numbers of complex checks may affect performance.
- ACL rules must be tested whenever record fields or user roles change.

6.5 Testing Summary

Test cases should include an authorized user, unauthorized user, administrator, matching field value, non-matching field value, and each CRUD operation. The expected result is that only requests satisfying the ACL rule are permitted.

6.6 Conclusion

The project demonstrates how programmable ACL rules can restrict access to individual records according to field values, improving security and supporting application-specific authorization requirements.